

Capstone Project Report on

FOLEO

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Foleo: Deterministic Tool Grounding for Reliable LLM-Based Portfolio Analytics

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Abstract—Large language models (LLMs) are increasingly used as conversational front-ends for personal investing, yet portfolio questions demand exact arithmetic and precisely defined risk metrics. This paper presents Foleo, a broker-integrated portfolio assistant in which an LLM handles dialogue while a deterministic tool layer computes every quantitative answer. Foleo maps broker holdings (Zerodha, Upstox) to exchange tickers, routes numeric requests—including implicit portfolio-level requests such as “what is my Sharpe ratio?”—to 16 tested functions, and answers them without invoking the LLM. We evaluate the design with a three-tier benchmark (portfolio arithmetic, single-stock time-series metrics, and portfolio risk metrics) on National Stock Exchange (NSE) prices pinned to a fixed snapshot, in which every system receives identical inputs and ground truth comes from an independent reference implementation. Against the open-weight reasoning model gpt-oss-120b, evaluated both as deployed in the application and with the complete data in its context, the tool path answered 26 of 26 cases exactly with zero LLM tokens and a few milliseconds of computation at most. The LLM was reliable on simple holdings arithmetic, but on time-series metrics it consumed 3,882 tokens per answer on average and reached 58.3% exact accuracy, with every missing answer caused by exhausting its token budget. Without price history, the deployed assistant gave no value in 47 of 48 requests, but once presented the portfolio’s gain since purchase as its three-month return. Two commercial assistants solved the task pack only by writing and executing their own code, and one of them implemented a different RSI formula than the one specified. We also document four failure modes uncovered while building the system—reasoning-budget exhaustion, ticker mis-resolution, metric-definition drift and evaluation artefacts—together with their fixes.

Index Terms—Large Language Models, Tool-Augmented Reasoning, Financial Question Answering, Portfolio Analytics, Numerical Reasoning, Risk Metrics, Hallucination, Evaluation Methodology.

I. INTRODUCTION

Conversational assistants built on large language models (LLMs) are an attractive interface for retail investors: a user can ask “how is my portfolio doing?” or “how risky is it?”

in plain language instead of navigating dashboards. Such questions, however, have exact answers. Invested value, profit and loss, a Sharpe ratio or a Value-at-Risk figure is either correct or wrong, and a plausible but wrong number is worse than no answer because it can drive a financial decision.

LLMs are known to be unreliable at exact multi-step computation. Their accuracy on compositional arithmetic decays quickly with problem size [1], and they can produce fluent statements that are not supported by their inputs [2]. Financial question-answering benchmarks confirm the gap in the domain: models trail human experts on numerical reasoning over financial reports [3]–[5], and on FinanceBench a retrieval-augmented GPT-4-Turbo answered incorrectly or refused 81% of questions [6]. A well-studied remedy is to let the model delegate computation to external tools or generated programs [7]–[11].

Most of this evidence comes from question-answering benchmarks rather than from deployed assistants connected to real brokerage accounts. In a deployed system, wrong numbers arise not only from arithmetic: holdings must be resolved to the correct securities, a metric such as RSI must be computed with one specific definition, the model may exhaust its generation budget, and the evaluation harness itself can report spurious agreement. This paper studies these issues end to end in Foleo, a portfolio assistant we built for Indian retail investors. Its contributions are:

- 1) **A tool-grounded assistant architecture.** Foleo separates conversation (LLM) from computation (deterministic tools). A rule-based router resolves broker holdings to exchange tickers, recognises implicit portfolio-level requests, and answers them from 16 deterministic functions without an LLM call.
- 2) **A controlled, reproducible benchmark.** Three task tiers are posed to every system with identical data: a pinned NSE price snapshot, explicit metric formulas, a fixed answer format and ground truth from an indepen-

dent reference implementation. Every raw response, the snapshot and the scoring code are retained in the project repository.

- 3) **Empirical findings** on accuracy, token cost, latency, run-to-run consistency and behaviour under missing data for a current open-weight reasoning model [12], complemented by two commercial assistants.
- 4) **A catalogue of failure modes** found during development, each with a fix and a regression test, including evaluation artefacts that had initially overstated the case for tools.

II. RELATED WORK

A. Numerical reasoning in LLMs

Chain-of-thought prompting improves multi-step reasoning by eliciting intermediate steps [13], but it leaves the arithmetic itself to token generation. Dziri et al. [1] show that transformers solve compositional tasks such as multi-digit multiplication largely by pattern matching over sub-computations, with accuracy falling sharply as problem size grows. Hallucination—content unsupported by the source—is a documented property of neural text generation [2].

B. Tool-augmented and program-aided models

Toolformer trains a model to call APIs, including a calculator, when helpful [7]. PAL and Program-of-Thoughts prompting have the model write a program that an interpreter executes, removing arithmetic from generation; PoT reports gains on the financial datasets FinQA, ConvFinQA and TAT-QA [8], [9]. ReAct interleaves reasoning with actions [10], and Mialon et al. [11] survey tool-augmented models. In these approaches the model still decides *what* to compute; Foleo instead fixes both the computation and the metric definition in tested code, and uses the LLM only where no tool applies.

C. Financial question answering and finance LLMs

FinQA [3], TAT-QA [4] and ConvFinQA [5] evaluate numerical reasoning over financial tables and text; FinanceBench [6] evaluates open-book QA over company filings. Domain models such as BloombergGPT [14] improve financial language understanding but are not designed to guarantee exact computation. Our benchmark differs in targeting *personal portfolio* analytics over broker holdings and price series, and in holding the input data identical across all systems.

D. Risk and performance metrics

Foleo implements standard definitions: the Sharpe ratio [15], historical Value-at-Risk [16] using the linear-interpolation sample quantile [17], maximum drawdown [18] and the Relative Strength Index [19]. Several of these have competing variants (e.g., Wilder’s smoothed RSI versus a simple moving average of gains and losses), which is why the formula must be stated explicitly in any comparison.

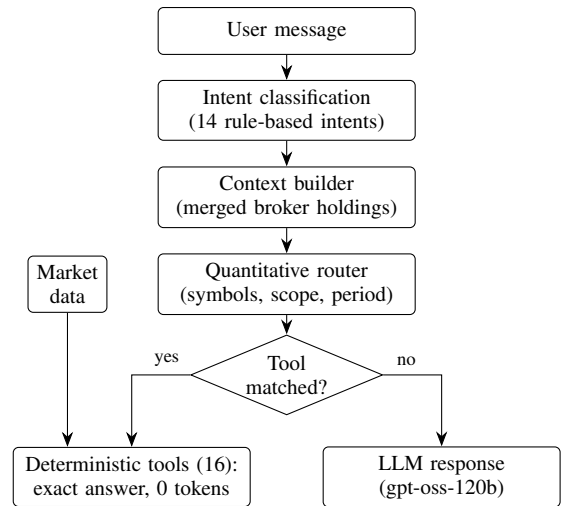


Fig. 1. Foleo’s conversational pipeline. Numeric requests that match a tool are answered deterministically; only the remaining requests reach the LLM.

III. SYSTEM DESIGN

A. Overview

Foleo consists of a FastAPI back end, a React (Vite) web client and a local SQLite store. Users link Zerodha (Kite Connect) and/or Upstox accounts through OAuth; holdings from all linked brokers are merged into one portfolio. Market data (daily prices and fundamentals) is obtained from Yahoo Finance through the `yfinance` library. The conversational layer uses an OpenAI-compatible chat API; the deployment evaluated here uses the open-weight reasoning model `gpt-oss-120b` [12] served by Groq.

B. Conversational pipeline

Fig. 1 shows the request flow. A rule-based classifier assigns one of 14 intents (e.g., portfolio summary, risk, rebalancing, goal tracking). A context builder assembles the merged holdings and per-holding statistics. The quantitative router then inspects the message; if it selects a tool and the tool succeeds, Foleo returns the tool’s formatted answer directly, records zero LLM tokens, and never calls the model. Otherwise the LLM answers with the portfolio context in its prompt. If the LLM is unavailable or returns an empty reply, a deterministic summary is returned instead of an empty message.

C. Symbol resolution and portfolio scope

Broker records identify a holding by trading symbol and exchange (e.g., `IDEA` on NSE), whereas the market data source expects an exchange-qualified ticker (`IDEA.NS`). Foleo maps each holding to its ticker using the exchange (`.NS` for NSE, `.BO` for BSE). Tokens in the user’s message are treated as securities only if they are a known alias (e.g., “gold”), carry an exchange suffix, match a held symbol, or are written in capitals; ordinary words such as “portfolio” are never treated as tickers. When a metric is requested and no security is named, the request applies to the whole portfolio.

For portfolio scope, Foleo evaluates the current holdings over the look-back window. With q_i the quantity of holding i and $P_{i,t}$ its closing price on trading day t (days common to all holdings),

$$V_t = \sum_i q_i P_{i,t}, \quad r_t = \frac{V_t}{V_{t-1}} - 1, \quad t = 1, \dots, T. \quad (1)$$

Single-stock metrics use $V_t = P_t$. Every metric function accepts either a ticker or a precomputed series, so stock-level and portfolio-level answers share one implementation.

D. Metric definitions

With $\bar{r}$ and s_r the sample mean and sample standard deviation of $\{r_t\}$, Foleo computes

$$R_{\text{total}} = 100 (V_T/V_0 - 1), \quad (2)$$

$$\text{MDD} = 100 \min_t \left(V_t / \max_{u \leq t} V_u - 1 \right), \quad (3)$$

$$\text{VaR}_{95} = -100 \hat{Q}_{0.05}(\{r_t\}), \quad (4)$$

$$\text{Sharpe} = \frac{\bar{r} - r_f / 252}{s_r} \sqrt{252}, \quad r_f = 0.05, \quad (5)$$

$$\text{RSI}_{14} = 100 - \frac{100}{1 + \bar{G}_{14} / \bar{L}_{14}}, \quad (6)$$

where $\hat{Q}_{0.05}$ is the linear-interpolation sample quantile [17] and $\bar{G}_{14}$, $\bar{L}_{14}$ are the simple means of the last 14 daily close-to-close gains and losses (not Wilder’s smoothed average [19]). Beta uses sample covariance and sample variance consistently. Holdings arithmetic uses invested value $I = \sum_i q_i a_i$ (with average buy price a_i), current value $C = \sum_i q_i p_i$ (with last price p_i), $\text{P\&L} = C - I$ and $\text{P\&L\%} = 100(C - I)/I$. For evaluation, a transparent classification rule maps $\text{P\&L\%}$ to Buy (≥ 10), Hold ($0 < x < 10$), Trim ($-10 < x \leq 0$) or Exit ($x \leq -10$), where x is the $\text{P\&L\%}$. The rule is illustrative and is not investment advice.

E. Routing and verification

The router is an ordered table of patterns (first match wins, so “value at risk” is matched before the generic “risk”), uses word boundaries so that, e.g., “various” does not trigger VaR, and extracts an explicit look-back period (1 month to 5 years) when stated; otherwise each metric keeps its own default window. The chatbot test suite (56 tests) passes, including a regression test for each failure mode in Section VI.

IV. EXPERIMENTAL METHODOLOGY

A. Research questions

- **RQ1 (accuracy):** Given identical data and explicit formulas, how often does an LLM reproduce the exact value that the deterministic tools produce?
- **RQ2 (cost and latency):** What token and time cost does each path incur?
- **RQ3 (missing data):** When the data needed for a metric is absent from its context, does the deployed assistant fabricate a value or decline?
- **RQ4 (external assistants):** Do general-purpose assistants with their own code execution match a fixed tool layer?

B. Data

Portfolios. P_1 is a three-holding snapshot of a real Upstox account (IDEA, YESBANK, SUZLON; one share each). P_2 is a synthetic eight-holding portfolio in the Upstox holdings schema with realistic quantities and prices (RELIANCE, TCS, HDFCBANK, INFY, ITC, TATAMOTORS, SBIN, ZOMATO); it is used only for arithmetic and is not a real account. *Prices.* Daily closes for six NSE securities (RELIANCE, TCS, HDFCBANK and the three P_1 holdings) were downloaded once for 25 June–25 September 2026 (67 trading rows), rounded to two decimals and pinned. The tools, the reference implementation, every LLM prompt and the external prompt pack all use this snapshot. Two rows (26 June and 14 September) repeat the previous day’s closes for all six securities, an artefact of the data source; because every system sees the same rows, the comparison is unaffected.

C. Tasks

The benchmark has 26 prompts. *Tier 1* (10 prompts) asks for invested value, current value, P&L, P&L% and the Buy/Hold/Trim/Exit class for P_1 and P_2 . *Tier 2* (12 prompts) asks for 3-month total return, maximum drawdown, 1-day 95% historical VaR and 14-day RSI for RELIANCE, TCS and HDFCBANK. *Tier 3* (4 prompts) asks for the Sharpe ratio, VaR, maximum drawdown and total return of portfolio P_1 , phrased implicitly (“my portfolio”) as a user would ask in the app. Every prompt states the formula in words and requires the reply to end with the line `FINAL: <value>`.

D. Systems compared

- **Foleo tools (ours):** the production router and tools, given the same question and holdings; the question is not rewritten for the tools.
- **Groq + full data:** openai/gpt-oss-120b called directly with a neutral system prompt containing all data the tools use (holdings and/or the price table); temperature 0, 4,096-token budget.
- **Groq chatbot (no prices):** Foleo’s LLM path as deployed (conversational system prompt, temperature 0.5, reasoning effort “low”, 1,284-token budget) with the raw holdings in context but, as in production, no price history. In production the context builder also precomputes per-holding P&L; the benchmark withholds these values so that the model must perform the arithmetic itself.
- **External assistants:** ChatGPT (reporting itself as GPT-5.6 Luna) and Claude Sonnet 5 (low effort), used through their web applications with the identical prompt pack.

E. Ground truth and scoring

Ground truth is computed by a separate pandas implementation of the formulas above, written independently of the tool code; the tools are scored against it like every other system. Only the `FINAL` line is parsed. A numeric answer is exact if it lies within ± 0.011 of the reference for Tier 1 (two-decimal rupee and percentage values), within ± 0.02 percentage points for returns, drawdown and VaR, within ± 0.1 for RSI and

within ± 0.005 for the Sharpe ratio; class labels must match exactly. A reply without a parseable `FINAL` line counts as “no answer”. Accuracy is reported with 95% Wilson intervals [20].

F. Protocol

Each Groq case was run 3 times per arm, with calls paced below the provider’s per-minute token limit. Calls rejected by the provider’s quota were not scored: they were logged separately (2 calls) and re-issued, and an interrupted run resumes from the same pinned snapshot, with a check that the ground truth is unchanged. Latency is client-side wall-clock time per answer; for the LLM it includes network and queueing time. Tool latency is computation time with market data already in memory; retrieving the data is a separate cost, measured at a median of 261 ms (95th percentile 4,859 ms) per security.

V. RESULTS

A. RQ1: Accuracy

Table I summarises the results. The tool path matched the reference on 26 of 26 cases and routed 10/10, 12/12 and 4/4 of the Tier 1, 2 and 3 prompts to the intended tool and scope, including the implicit portfolio-level questions of Tier 3.

On Tier 1, with the holdings in its context, the LLM was also reliable: 100.0% exact with full data and 96.7% through the deployed chatbot. Simple holdings arithmetic is therefore not where the LLM fails. On the time-series tiers, where the answer requires processing a 67-row price table, accuracy with full data fell to 58.3% (Tier 2) and 25.0% (Tier 3). Of the 15 Tier 2 misses, 14 ended without a final answer (14 of them by exhausting the 4,096-token budget, typically while sorting daily returns to obtain the VaR quantile) and 1 gave a wrong value; for Tier 3 the corresponding counts are 9, 9 (9 truncated) and 0.

Accuracy depended strongly on the metric (Table III). On Tier 2 the full-data LLM was exact on all 9 total-return answers, 7 of 9 drawdown answers and 5 of 9 RSI answers, but on none of the 9 VaR answers, all of which ran out of budget. On Tier 3 it solved only the portfolio total return; every Sharpe, VaR and drawdown attempt was truncated. The single wrong value is instructive: for the HDFCBANK RSI, one repeat mis-added the daily gains and losses and reported 62.68 against the reference 61.40, with correct-looking working, while the other repeats of the same prompt at temperature 0 behaved differently. Such an error is invisible to a user.

B. RQ2: Cost and latency

The tool path uses no LLM tokens and computed answers in a median of 0.120 ms (Tier 1) and 0.855 ms (Tier 2); portfolio metrics (Tier 3) took 3.298 ms because the holdings’ price series are first combined. The full-data LLM used 1,421 tokens per Tier 1 answer and 3,882 per Tier 2 answer on average (including hidden reasoning tokens), with median latencies of 2,481.5 ms and 8,236.9 ms. The deployed chatbot carries a long conversational system prompt, so even its Tier 1 answers cost 2,694 tokens on average. Across repeats, the full-data

LLM gave the identical answer on every repeat in 10/10 Tier 1 and 6/12 Tier 2 cases; the tools are deterministic by construction.

C. RQ3: Behaviour without price data

Given holdings but no price history, the deployed chatbot gave no final value in 47 of 48 time-series requests (100.0% of Tier 2, 91.7% of Tier 3). In these replies the model stated that it lacked the historical prices (e.g., “I don’t have the price from three months ago”). In the remaining 1 reply, asked for the portfolio’s three-month total return, it answered: “Your portfolio slipped from Rs.89 (initial) to Rs.83 (last), a -6.74% three-month return.” The figures are the invested and current values of the holdings, so -6.74% is the gain since purchase; the actual three-month return is -19.27% . The model substituted a quantity it could compute for the one that was asked, and presented it with the requested label. The other two repeats of the same prompt declined. Refusal is therefore the common behaviour, but not a guarantee: a user cannot tell which of the two they received. The tool path avoids both outcomes by fetching the data itself.

D. RQ4: External assistants

Both assistants received the whole prompt pack in a single session and chose to solve it by writing and executing code rather than by generating the arithmetic token by token. Claude answered 26/26 prompts exactly. ChatGPT answered 23/26: its three errors are all RSI values (e.g., 26.22 against the reference 25.66 for RELIANCE), although the prompt specified the simple-average formula. We could not reproduce its values with common RSI variants (Wilder smoothing, 13- or 15-day windows, excluding unchanged days), so the model implemented a definition of its own. Code execution removes arithmetic error but not *definition* error: when the model writes the program, it also chooses the formula. Because these runs used consumer interfaces with code execution enabled, a single session and no timing, we report them as qualitative evidence only (Table II).

VI. FAILURE MODES FOUND DURING DEVELOPMENT

Building and evaluating Foleo surfaced four classes of failure. Each was fixed and is covered by a regression test.

1) Reasoning-budget exhaustion. The deployed chatbot originally capped generation at 260 tokens. For a reasoning model, hidden reasoning tokens count against the same budget: in a pre-fix run, all 18 of 18 chatbot replies were empty, each having consumed exactly 260 completion tokens. Users would have received blank answers. The fix sets a low reasoning effort, adds headroom to the budget, and replaces any empty reply with a deterministic summary.

2) Ticker mis-resolution. The original symbol extractor treated ordinary words as tickers (“portfolio”, “this”, “maximum”) and passed broker symbols without an exchange suffix. The bare symbol `IDEA` resolves on the data source to an unrelated US-listed security (last close \$0.02) rather than Vodafone Idea (`IDEA.NS`, last close Rs. 14.26), and “Sharpe

TABLE I

ACCURACY, COST AND LATENCY BY TASK TIER AND SYSTEM ARM. ACCURACY IS EXACT AGREEMENT WITH THE INDEPENDENT REFERENCE WITHIN TOLERANCE; 95% WILSON INTERVALS IN BRACKETS. LATENCY IS WALL-CLOCK PER ANSWER; TOKENS ARE LLM TOKENS PER ANSWER (PROMPT + COMPLETION, INCLUDING HIDDEN REASONING).

Tier	Arm	n	Exact acc. (%) [95% CI]	No answer (%)	Median lat. (ms)	p95 lat. (ms)	Mean tokens
Portfolio arithmetic	Foleo tools (ours)	10	100.0 [72.2, 100.0]	0.0	0.120	1.413	0
	Groq + full data	30	100.0 [88.6, 100.0]	0.0	2,481.5	7,297.8	1,421
	Groq chatbot (no prices)	30	96.7 [83.3, 99.4]	3.3	1,644.4	3,823.6	2,694
Stock time-series metrics	Foleo tools (ours)	12	100.0 [75.7, 100.0]	0.0	0.855	2.488	0
	Groq + full data	36	58.3 [42.2, 72.9]	38.9	8,236.9	14,683.4	3,882
	Groq chatbot (no prices)	36	0.0 [0.0, 9.6]	100.0	1,245.9	7,539.7	2,519
Portfolio risk metrics	Foleo tools (ours)	4	100.0 [51.0, 100.0]	0.0	3.298	5.055	0
	Groq + full data	12	25.0 [8.9, 53.2]	75.0	9,715.3	13,575.9	4,867
	Groq chatbot (no prices)	12	0.0 [0.0, 24.3]	91.7	1,917.8	18,553.2	2,285

TABLE II

EXTERNAL ASSISTANTS ON THE IDENTICAL PROMPT PACK (SINGLE SESSION, BUILT-IN CODE EXECUTION USED BY BOTH). EXACT ANSWERS / PROMPTS.

Assistant	Tier 1	Tier 2	Tier 3	Total
ChatGPT (GPT-5.6 Luna)	10/10	9/12	4/4	23/26
Claude Sonnet 5 (low)	10/10	12/12	4/4	26/26

ratio of my portfolio” was computed for the first holding only. In our initial six-prompt suite, four prompts failed for this reason, and their reported “tool latencies” were in fact the time taken to receive an error. The fix is exchange-aware mapping and explicit portfolio scope (Section III-C).

3) Metric and routing defects. Beta divided a sample covariance by a population variance, inflating every beta by a factor $n/(n-1)$; the VaR keyword matched inside unrelated words; and an unstated period silently overrode each metric’s default window. All three are fixed.

4) Evaluation artefacts. An early harness scored the LLM by comparing the *first* number in each reply with the tool’s value. Numbers echoed from the prompt (e.g., “6-month”) produced false matches: 3 of 11 reported agreements were spurious, and the true agreement was 0 of 11. That harness also compared a data-rich tool against an LLM given no price data. Both problems overstated the advantage of tools; the protocol of Section IV (identical data, stated formulas, a mandatory FINAL line and an independent reference) was designed to remove them.

VII. DISCUSSION

A. Why ground the numbers?

The case for deterministic tools does not rest on the LLM being unable to add: on holdings arithmetic it was reliable. It rests on five properties the tools provide and the LLM does not: (i) *completeness*—tool answers do not disappear when a token budget runs out; (ii) *cost*—zero tokens against thousands per answer; (iii) *determinism*—the same question always yields the same number, whereas the LLM’s answers varied across repeats and were occasionally, silently wrong; (iv) *definitional stability*—a metric always means the tested

formula, which even code-writing assistants did not guarantee; and (v) *auditability*—every figure can be traced to a function, a data snapshot and a test. Wrong values were rare in our runs (one arithmetic error and one substituted quantity), but they are the failures a user cannot detect. The LLM remains valuable for intent understanding, explanation and the long tail of questions that no tool covers.

B. Implications for evaluation

Our own early measurements overstated the benefit of tools through first-number scoring and unequal data access (Section VI). Comparisons between LLMs and tools in finance should give every system the same data, state the metric definition, fix the answer format, and compute ground truth independently of the system under test.

C. Threats to validity

The benchmark is small (26 prompts, 3 repeats per LLM arm) and uses one open-weight model through one provider, so the confidence intervals are wide. Portfolio P_2 is synthetic. The price window is three months of daily data and contains two stale rows from the data source. The external assistants were run once, in a single session, through consumer interfaces with code execution enabled. Latencies were measured on a shared free-tier service and vary with load. Agreement between the tools and the reference implementation establishes that the implementation matches the stated definitions, not that the definitions are the right ones for a given investor. The Buy/Hold/Trim/Exit rule is a fixed test rule, not a recommendation, and nothing in this work is investment advice.

VIII. CONCLUSION

Foleo shows that a portfolio assistant can combine the fluency of an LLM with exact, reproducible numbers by routing every quantitative request—including implicit portfolio-level ones—to deterministic, tested tools. In a controlled comparison with identical inputs, the tools were exact at zero token cost, while a current reasoning model was reliable on simple arithmetic but costly, often incomplete and occasionally silently wrong on time-series risk metrics, and code-writing assistants still drifted in metric definitions. The main lesson from building the system is that reliability depends as much

on symbol resolution, metric definitions, generation budgets and evaluation design as on arithmetic.

IX. FUTURE WORK

- Extend the benchmark to more prompts, longer price histories and several model families, including an arm in which the LLM invokes Foleo’s tools through function calling instead of the rule-based router.
- Evaluate the explanatory layer with human raters: whether LLM explanations restate tool results faithfully.
- Integrate the retrieval index already present in the code base (a sentence-embedding vector store of financial concepts) into the conversational pipeline, and measure its effect.
- Add conformance tests that pin each metric to published reference values, and expose the definition used alongside every answer.

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APPENDIX A

PROMPT TEMPLATE AND PER-CASE RESULTS

Every benchmark prompt has the form below; the full pack of 26 prompts, the price snapshot, the ground truth and all raw model responses are included in the project repository.

```
You are a precise financial analyst. Use
only the data provided and compute exactly.
<holdings JSON and/or daily closing prices
CSV>
<question> <formula stated in words>
Show brief working, then end with one final
line exactly in the form: FINAL: <number>
```

Table III lists every answer on the time-series tiers.

TABLE III
PER-CASE ANSWERS ON THE TIME-SERIES TIERS (ALL REPEATS). “-” = NO FINAL NUMERIC ANSWER.

Case	Reference	Foleo tools	Groq + full data	Groq chatbot (no prices)
RELIANCE.NS:total_return	-6.99	-6.99	-6.99, -6.99, -6.99	-, -, -
RELIANCE.NS:max_drawdown	-8.66	-8.66	-8.66, -8.66, -8.65	-, -, -
RELIANCE.NS:var95	1.69	1.69	-, -, -	-, -, -
RELIANCE.NS:rsi14	25.66	25.66	25.65, -, -	-, -, -
TCS.NS:total_return	-0.61	-0.61	-0.61, -0.61, -0.61	-, -, -
TCS.NS:max_drawdown	-15.83	-15.83	-, -15.85, -15.84	-, -, -
TCS.NS:var95	2.75	2.75	-, -, -	-, -, -
TCS.NS:rsi14	22.21	22.21	22.22, -, 22.22	-, -, -
HDFCBANK.NS:total_return	-7.62	-7.62	-7.62, -7.62, -7.62	-, -, -
HDFCBANK.NS:max_drawdown	-17.2	-17.2	-, -17.2, -17.2	-, -, -
HDFCBANK.NS:var95	2.19	2.19	-, -, -	-, -, -
HDFCBANK.NS:rsi14	61.4	61.4	61.38, 62.68, 61.4	-, -, -
portfolio:sharpe	-4.2766	-4.2766	-, -, -	-, -, -
portfolio:var95	2.38	2.38	-, -, -	-, -, -
portfolio:max_drawdown	-20.77	-20.77	-, -, -	-, -, -
portfolio:total_return	-19.27	-19.27	-19.27, -19.27, -19.27	-6.74, -, -